Term review and consolidation

Lecture and worked examples

Maths Explorer

Individual review

Work without the course notes for 25 minutes, then check with the methods in the booklet. For each question, note: solved alone, solved with help, or to redo. Justify the answers.

Individual review

1. Reduce $90/126$ to lowest terms. Prove that the sum of two odd integers is even.
2. With $A(-1, 2)$ and $B(3, -2)$, calculate $\overrightarrow{AB}$, AB and the midpoint of $[AB]$ in an orthonormal frame.
3. Determine an equation of (AB); does $C(5, -4)$ belong to this line?
4. Calculate $[-2, 4] \cap (1, 7]$ and $[-2, 4] \cup (1, 7]$.
5. Solve $|x - 1| \leq 4$ and explain why $1/3 \notin \mathbb{D}$.

Worked answer: Individual review

1. $\gcd(90, 126) = 18$, so $90/126 = 5/7$. If $u = 2k + 1$ and $v = 2\ell + 1$, then $u + v = 2(k + \ell + 1)$ is even.

2. $\overrightarrow{AB} = (4, -4)$; $AB = 4\sqrt{2}$; midpoint $I(1, 0)$.

3. $y = -x + 1$. For C, $-4 = -5 + 1$: it belongs to (AB).

4. Intersection $(1, 4]$; union $[-2, 7]$.

5. $x \in [-3, 5]$. Writing $1/3 = a/10^n$ would require 10^n to be divisible by 3, which is impossible.

Correcting an error and revisiting a method

Correct: (a) "$\overrightarrow{AB} = (x_A - x_B, y_A - y_B)$"; (b) "The zero vector directs every line"; (c) "$|x - 2| \leq 3$ gives $x \in [2, 3]$"; (d) "$1/3 \approx 0.333$, so $1/3$ is decimal".

Worked answer: Correcting an error and revisiting a method

(a) Subtract the origin from the endpoint: $(x_B - x_A, y_B - y_A)$. (b) A direction vector must be nonzero. (c) The distance to 2 is at most 3: $x \in [-1, 5]$. (d) 0.333 differs from 1/3: $3 \times 0.333 = 0.999 \neq 1$. A decimal approximation does not determine the nature of the approximated number.

Correcting an error and revisiting a method

1. Reduce $72/108$, then $\frac{3}{4} - \frac{1}{6}$.
2. With $P(1,1)$ and $Q(5,3)$, determine the midpoint, a direction vector and an equation of (PQ).
3. Determine $(-\infty, 3] \cap [-1, +\infty)$, then solve $|x+2| < 1$.

Worked answer: Correcting an error and revisiting a method

1. $2/3$; $7/12$. 2. Midpoint $(3, 2)$, direction vector $(2, 1)$, equation $x - 2y + 1 = 0$. 3. Intersection $[-1, 3]$; solution of the inequality $(-3, -1)$.

Applying and self-assessing

Two displacements are represented by $\vec{u}(6, 4)$ and $\vec{v}(9, 6)$.

Applying and self-assessing (2)

1. Show that they are collinear and find the coefficient relating $\vec{v}$ to $\vec{u}$.
2. The starting point is $A(-2, 1)$. Determine the endpoints B and C associated with the displacements $\vec{u}$ and $\vec{v}$.
3. Find an equation of the line through these three points.
4. A point $M(x, y)$ of this line must satisfy $|x - 4| \leq 3$. Give the interval of possible abscissae.

Worked answer: Applying and self-assessing

1. $\vec{v} = \frac{3}{2}\vec{u}$. 2. $B(4, 5)$ and $C(7, 7)$. 3. A direction vector is $(3, 2)$; the equation is $2x - 3y + 7 = 0$, satisfied by the three points. 4. $|x - 4| \leq 3 \iff x \in [1, 7]$.

Applying and self-assessing

My priority:

The mistake to avoid:

The method to reuse:

An exercise to redo without help: